MAT 183 Chapter 6.3 - 6.4 Practice Problems Part-2

Name_____

Central Limit Theorem:

1. A bank's loan officer rates applicants for credit. The ratings are normally distributed with a mean of 200 and a standard deviation of 50. If 40 different applicants are randomly selected, find the probability that their mean is above 215. 1) _____

A) 0.3821 B) 0.1179 C) 0.4713 D) 0.0287

2. A study of the amount of time it takes a mechanic to rebuild the transmission for a 2005 Chevrolet Cavalier shows that the mean is 8.4 hours and the standard deviation is 1.8 hours. If 40 mechanics are randomly selected, find the probability that their mean rebuild time exceeds 8.7 hours. 2) _____

A) 0.1469 B) 0.1285 C) 0.1346 D) 0.1946

3. A final exam in Math 160 has a mean of 73 with standard deviation 7.8. If 24 students are randomly selected, find the probability that the mean of their test scores is greater than 71.

Sampling Distributions and Estimators:

4. Personal phone calls received in the last three days by a new employee were 3, 2, and 5. Assume that samples of size 2 are randomly selected with replacement from this population of three values. List the different possible samples and find the proportion of even number for each of them.

Sample									
Proportion of Even #									

5. Three randomly selected households are surveyed as a pilot project for a larger survey to be conducted later. The numbers of people in the households are 3, 4, and 7. Consider the values of 3, 4, and 7 to be a population. Assume that samples of size n = 2 are randomly selected with replacement from the population of 3, 4, and 7. The nine different samples are as follows: (3,3) $(3,4)$, $(3,7)$, $(4,3)$, $(4,4)$, $(4,7)$, $(7,3)$, $(7,4)$, and $(7,7)$.

(i) Construct a probability distribution table that describes the sampling distribution of the proportion of even numbers when samples of size n = 2 are randomly selected.

(ii) Does the mean of the sample proportions target the value of the population proportion? Yes it target the value of the population proportion

(iii) Does the sample proportion make a good estimator of the population proportion? Yes

Sample									
Proportion of even#									

i) Probability Distribution Table:

Proportion			
Probability Distribution P(x)			

ii) Mean of the Sample Proportion =

Population Proportion of Even number = _____

Comparing Mean of Sample Proportion to population proportion: _____.

iii) Since the Sample proportion ----------the value of the population proportion. So, the sample proportion is an ---------- estimator.

6. Flood insurance policies sold in the last three days by a new agent were 3, 5, and 7. Assume that samples of size 2 are randomly selected with replacement from this population of three values. List the different possible samples and find the median of each of them.

Try yourself:

7. Three randomly selected households are surveyed as a pilot project for a larger survey to be conducted later. The numbers of people in the households are 2, 3, and 9. Consider the values of 2, 3, and 9 to be a population. Assume that samples of size n = 2 are randomly selected with replacement from the population of 2, 3, and 9. The nine different samples are as follows: $(2, 2)$, $(2, 3)$, $(2, 9)$, $(3, 2)$, $(3, 3)$, $(3, 9)$, $(9, 2)$, $(9, 3)$, and $(9, 9)$.

(i) Find the mean of each of the nine samples, then summarize the sampling distribution of the means in the format of a table representing the probability distribution.

(ii) Compare the population mean to the mean of the sample means.

(iii) Do the sample means target the value of the population mean? In general, do means make good estimators of population means? Why or why not?

Solution:

Sample									
Sample Mean									

i) Probability Distribution Table:

Sample Mean						
Probability Distribution P(x)						

ii) Mean of the Sample mean = _____

Population Mean = (_____

iii) Comparing mean of sample mean and population mean, we saw that they _____ to each other.

Since the Sample mean _____the value of the population mean. So, the sample mean is a ____ estimator.

8. Three randomly selected households are surveyed as a pilot project for a larger survey to be conducted later. The numbers of people in the households are 2, 4, and 8. Consider the values

of 2, 4, and 8 to be a population. Assume that samples of size n = 2 are randomly selected with replacement from the population of 2, 4, and 8. The nine different samples are as follows: $(2, 2)$, $(2, 4)$, $(2, 8)$, $(4, 2)$, $(4, 4)$, $(4, 8)$, $(8, 2)$, $(8, 4)$, and $(8, 8)$.

(i) Find the mean of each of the nine samples, then summarize the sampling distribution of the means in the format of a table representing the probability distribution.

(ii) Compare the population mean to the mean of the sample means.

(iii) Do the sample means target the value of the population mean? In general, do means make good estimators of population means? Why or why not?

9. Four randomly selected households are surveyed as a pilot project for a larger survey to be conducted later. The numbers of people in the households are 2, 3, 6, and 7. Consider the values of 2, 3, 6, and 7 to be a population. Assume that samples of size n = 2 are randomly selected with replacement from the population of 2, 3, 6, and 7. List the sixteen different samples and find the proportion of even number for each of them.